Faculty of Computing

BSc (Hons) in Software Engineering
Year 2 Semester 2 (2026)

SE2020 - Web and Mobile Technologies Lab 09

Lab Sheet – React Basics

Objectives

At the end of this lab, students will be able to:

- Understand what React is
- Create a React application using Vite
- Understand project structure
- Create and use React components
- Use JSX syntax
- Pass data using props
- Handle basic events

Prerequisites

- Node.js 18+ installed
- Basic JavaScript knowledge
- A code editor (VS Code)

Task 1: Check Node.js Installation

1. Install node js to laptop
   Link - `https://nodejs.org/en/download`

2. Open Terminal / Command Prompt
3. Run the following command:
   Windows OS - node -v
   Mac OS - node -v

If Node.js is installed, the version number will be displayed.

Task 2: Create React App Using Vite

1. Open Terminal
2. Run the following command

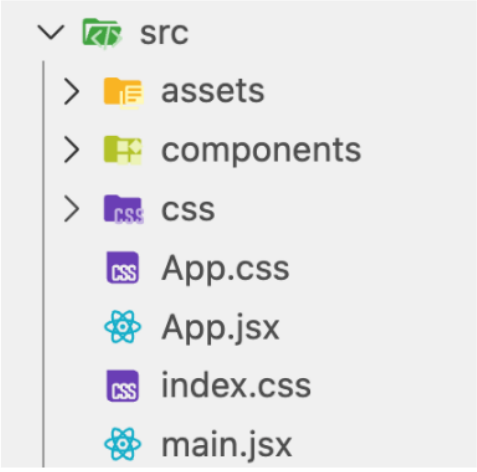
```
npm create vite@latest react-basic-lab
```

3. Navigate to the folder

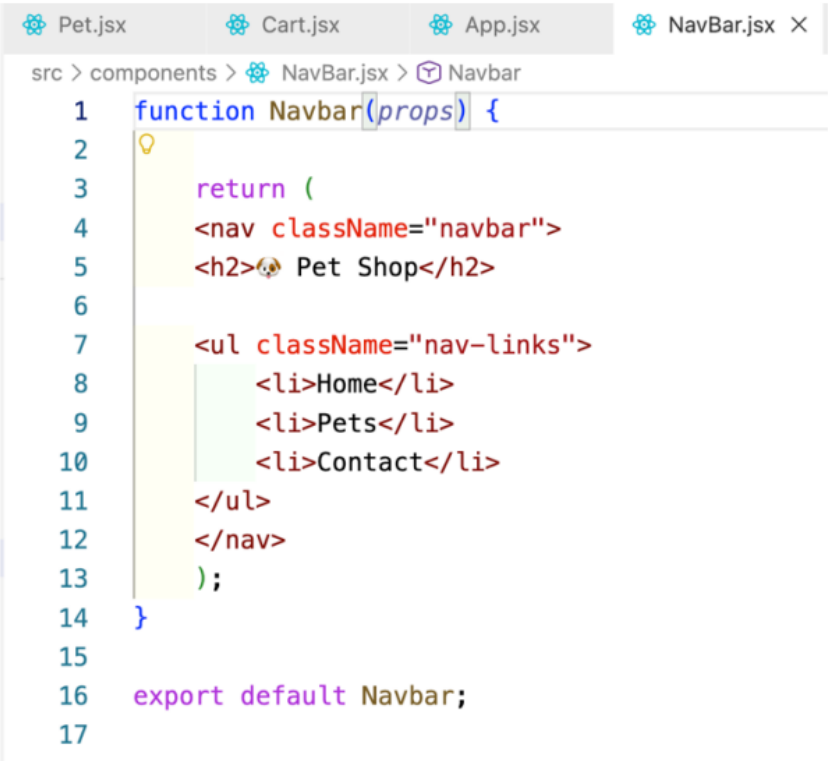
   cd latest react-basic-lab

4. Create the project structure as follows.

   Inside the 'src' folder create the 'components' folder.



Inside the component folder create three files called 'NavBar.jsx', 'Pet.jsx' and 'Cart.jsx'.
Inside the ' NavBar.jsx ' type this code.



Then inside the ' Cart.jsx ' add this code.

Pet.jsx | Cart.jsx | App.jsx | NavBar.jsx

src > components > Cart.jsx > Cart

```jsx
function Cart() {

return (
    <div className="cart">
    <h2>🛒 Shopping Cart</h2>
    <p>Total Items in Cart:</p>
    </div>
);
}

export default Cart;
```

Finally add below code inside the ‘ Pet.jsx ’ .

Pet.jsx | Cart.jsx | App.jsx | main.jsx

src > components > Pet.jsx > Pet

```jsx
function Pet(props) {
  return (
    <div className="pet-card">
      <h3>Web Name</h3>
      <p>Type: </p>
      <p>Price: Rs. </p>
      <p>Stock: </p>

      <button >
        Add to Cart
      </button>
    </div>
  );
}

export default Pet;
```

5. Replace all the content of the App.jsx file with the code below

App.jsx

src > App.jsx > ...

```jsx
import './App.css'

function App() {
  return (
    <div className="App">
      <h1>Hello World!</h1>
    </div>
  );
}

export default App;
```

Task 3: Component

1. Inside the 'App.jsx' file, call the 'Navbar','Pet", and 'Cart' files as follows and import them as below.
2. Then run and see the output.
3. Inside the 'App.jsx' file call the student file as follows and import it as below.

App.jsx × main.jsx NavBar.jsx 1 index.css styles.css

App.jsx > ...

```
import Navbar from "./components/Navbar";
import Pet from "./components/Pet";
import Cart from "./components/Cart";

function App() {

  return (
    <div>
      <Navbar />

      <div className="container">
        <h1>🐾 Welcome to Pet Shop</h1>
        <p>Check pet availability and add to cart!</p>

        <Pet />

        <Pet />

        <Pet/>

        <Cart />
      </div>
    </div>
  );
}

export default App;
```

4. Create folder called 'css' inside the 'src' folder. Inside that one create a file call 'styles.css'. Copy past the code which included inside the zip folder. 'styles.css'.

Task 4: Usage of props

1. Inside the NavBar.jsx add the below code

```
<li>🛒 Cart ({props.cartCount})</li>
```

Explanation:

- props.cartCount passes **cart count from App** to Navbar
- Props allow **data transfer from parent to child**

2. Inside the 'Pet.jsx' file add below code

Pet.jsx × Cart.jsx App.jsx main.jsx

src > components > Pet.jsx > Pet

```
function Pet() {
  return (
    <div className="pet-card">
      <h3>Web Name</h3>
      <p>Type: </p>
      <p>Price: Rs. </p>
      <p>Stock: </p>

      <button >
        Add to Cart
      </button>
    </div>
  );
}

export default Pet;
```

3. Add the below code inside the 'App.js' file

```
<Pet
  name="Golden Retriever"
  type="Dog"
  price={50000}
/>

<Pet
  name="Persian Cat"
  type="Cat"
  price={35000}
/>

<Pet
  name="Parrot"
  type="Bird"
  price={15000}
/>
```

Task 5: React Hooks

1. In this scenario we implement **add to cart function, decrease stock count and cart count in navigation bar.**

   Firstly create the two use state .One for cart count other one for stock count.

   At the initially we assign the cart count as zero. There for inside the useState bracket we input 0 . And the initial values for the stock we need to add a count like as follows.

```
const [cartCount, setCartCount] = useState(0); // useState for cart

const [stock, setStock] = useState({           // useState for pet stock
  "Golden Retriever": 5,
  "Persian Cat": 3,
  "Parrot": 10
});
```

2. Then we need to create the function call addToCart () . Inside that function need to be included followings.

```
function addToCart(petName) {                  // Event handler
  if (stock[petName] > 0) {
    setCartCount(cartCount + 1);
    setStock({ ...stock, [petName]: stock[petName] - 1 });
  } else {
    alert(`${petName} is out of stock!`);
  }
}
```

   Using useEffect() Hook

3. Add below code inside the 'App.jsx file'

```
// useEffect for side effects
useEffect(() => {
  console.log("Cart count:", cartCount);
  console.log("Current stock:", stock);
}, [cartCount, stock]);
```

   Inside the console see what happen.

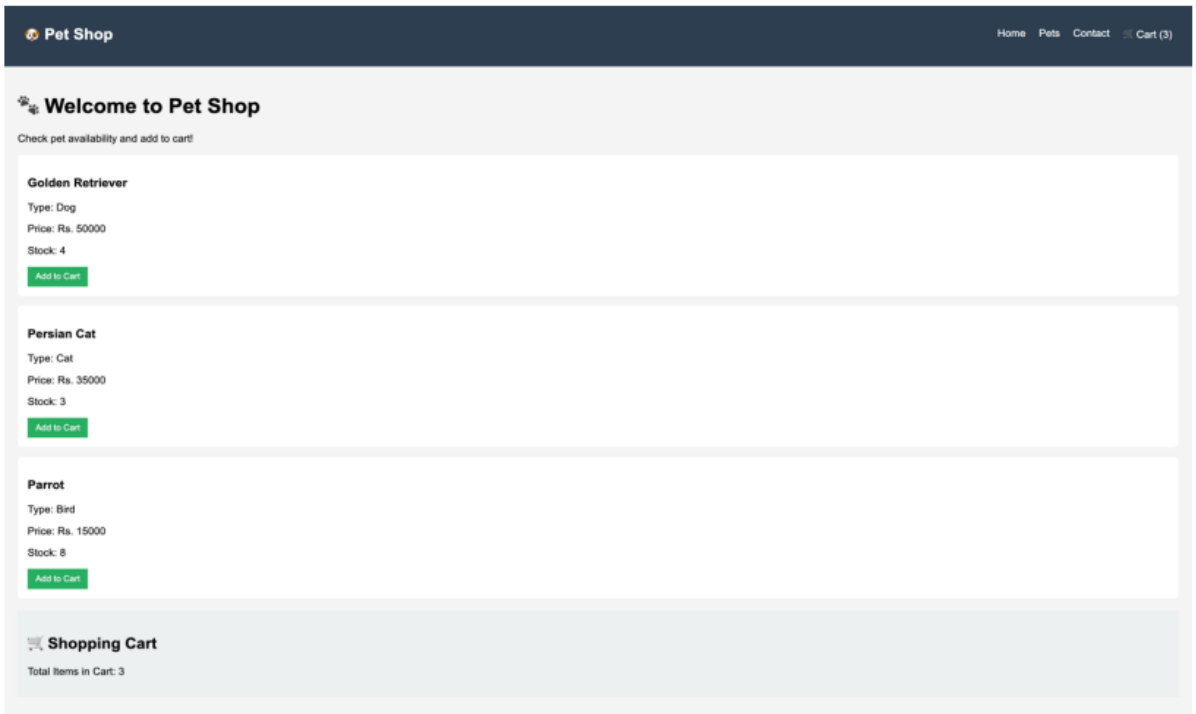
Explanation: We use useEffect here to run extra code after the component updates, and what happens is that whenever the cart count or stock changes, React automatically

runs this code and logs the updated values to the console, helping us track changes or perform side tasks like logging or API calls.

Task 6: Handling Events

We use the onClick event handler inside the button so that when the user clicks the button, React executes the given function (like adding a pet to the cart), allowing the app to respond to user actions.

```
<button onClick={props.onAddToCart} disabled={props.stock === 0}>
  {props.stock > 0 ? "Add to Cart" : "Out of Stock"}
</button>
```

Explanation: This button calls the add to cart function when clicked, and it automatically disables itself and shows "Out of Stock" when the stock becomes zero, otherwise it shows "Add to Cart".

Final Output: